IoT-Based Smart Water Level Monitoring and Billing System

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***Abstract*—Water resource management is increasingly challenging due to urbanization and inefficient monitoring systems, leading to overflow, wastage, and lack of consumption awareness. This paper presents an IoT-based smart water level detection, monitoring, and billing system designed using a cloud-integrated architecture. The system utilizes an ultrasonic sensor for water level detection and a Hall-effect flow sensor for accurate water usage measurement. An ESP32 microcontroller processes the data, controls a water pump using predefined thresholds, and communicates with the Blynk IoT cloud platform for near real-time monitoring and user interaction over Wi-Fi. The system integrates monitoring, automation, and billing within a unified cloud-based framework. A billing mechanism is implemented using a pulse-to-liter conversion model to estimate water consumption cost. Experimental results demonstrate reliable system performance, low latency, and effective automation. The proposed solution is cost-effective, efficient, and suitable for smart water management in residential and small-scale applications.**

***Index Terms*—IoT, ESP32, Water Monitoring, Flow Sensor, Smart Billing, Blynk IoT, Cloud Monitoring, Automation**

I. Introduction

Water is one of the most essential natural resources, yet its efficient management remains a major challenge in residential, commercial, and small-scale industrial environments. Traditional water management systems rely heavily on manual monitoring or basic automation, which often results in inefficiencies such as overflow, delayed pump operation, and lack of awareness regarding consumption patterns. These issues lead to unnecessary water wastage and increased operational cost.

The rapid advancement of the Internet of Things (IoT) has enabled the development of intelligent monitoring and control systems. IoT integrates sensors, embedded systems, and communication technologies to provide near real-time data acquisition, processing, and decision-making.

Despite these advancements, existing IoT-based water monitoring systems primarily focus on water level detection and pump automation. They lack accurate water consumption measurement and billing mechanisms, which limits their applicability in real-world scenarios, especially in shared environments.

A. Challenges and Research Gaps

The key challenges identified in existing systems include:

- Absence of accurate water usage measurement
- Lack of billing integration based on consumption
- Dependence on cloud infrastructure without optimized near real-time performance
- Limited near real-time visualization in local environments

B. Objectives and Contributions

The main contributions of this work are:

- Design of a near real-time IoT-based water monitoring system
- Integration of flow sensor for usage measurement
- Implementation of a billing system based on consumption
- Development of a cloud-based monitoring interface using Blynk IoT

C. Organization of the Paper

The rest of the paper is organized as follows: Section II presents the literature review, Section III describes the proposed system architecture, Section IV discusses implementation and experimental setup, Section V presents results and discussion, and Section VI concludes the paper.

II. Literature Review

A. Overview of Existing IoT Applications in Water Monitoring

The rapid advancement of Internet of Things (IoT) technology has significantly improved water monitoring and management systems. IoT enables the integration of sensors, microcontrollers, and cloud platforms to collect and transmit near real-time data, which helps optimize water usage and reduce wastage.

Several studies have explored IoT-based water level monitoring systems for residential and industrial applications. An

IoT-enabled water monitoring system using ultrasonic sensors and ESP modules was proposed to measure tank water levels and provide visual alerts when the water exceeds predefined thresholds. The system also provides near real-time graphical visualization through a cloud-based platform, allowing users to monitor tank levels remotely [1].

Another study presented a web-based IoT monitoring system that integrates ultrasonic flow meters with an ESP32 microcontroller to measure water flow rate and consumption. The system transmits near real-time flow data to a cloud database and visualizes the information through a web-based dashboard. Experimental results showed that the ultrasonic flow meter achieved a measurement accuracy of approximately $\pm 1\%$ compared to conventional flow meters [2].

In addition to monitoring water flow, some research focuses on water consumption tracking and billing estimation. An IoT-based water bill monitoring system was developed using a NodeMCU ESP8266 microcontroller and a water flow sensor. The system measures water usage and transmits the data to the ThingSpeak cloud platform for near real-time monitoring. The billing amount is automatically calculated based on the measured water consumption and predefined tariff rates [3].

Furthermore, IoT-based near real-time water level monitoring systems using ESP32 microcontrollers have been proposed for critical infrastructure applications such as reservoirs, dams, and storage tanks. These systems utilize ultrasonic sensors to measure tank levels and transmit data through wireless communication networks for remote monitoring and early warning of abnormal water levels [4].

In a comprehensive review of IoT-based smart water management systems, it was highlighted that IoT technologies, wireless sensor networks, and cloud computing play an important role in improving water distribution efficiency, monitoring water quality, and reducing water wastage in agriculture and urban environments [5].

B. Discussion of Previous Approaches and Their Limitations

Although many IoT-based water monitoring systems have been developed, several limitations still exist in current approaches.

Many existing systems focus primarily on water level monitoring using ultrasonic or float sensors. While these systems effectively detect water levels and prevent tank overflow, they often lack mechanisms to measure actual water consumption. As a result, they cannot provide insights into water usage patterns or consumption trends [1].

Some studies have incorporated flow measurement sensors to monitor water discharge. Although these systems provide more accurate water usage measurements, they may require complex sensor calibration or specialized hardware components, which increases system cost and implementation complexity [2].

IoT-based water billing systems have also been proposed, where flow sensor data is used to calculate water bills in near real-time. However, such systems depend heavily on reliable network connectivity and cloud services. Network instability can affect near real-time data transmission and reduce the reliability of monitoring and billing updates [3].

In addition, several monitoring systems emphasize automation and remote monitoring but do not address security and scalability issues. Since water management infrastructure can be considered critical infrastructure, ensuring secure data communication and protection against cyber threats is essential. Some recent studies have highlighted the importance of secure communication protocols and encryption mechanisms in IoT monitoring systems [4].

Moreover, existing smart water management solutions often focus on specific applications such as irrigation monitoring, water quality detection, or tank level monitoring individually rather than integrating multiple functionalities within a single system [5].

C. Summary of Research Gaps

Based on the analysis of existing studies, several research gaps can be identified in current IoT-based water monitoring systems.

First, many systems focus only on water level monitoring and do not incorporate direct water consumption measurement using flow sensors. This limits the ability to track actual water usage.

Second, although some systems measure water flow, they often do not integrate automatic billing mechanisms that estimate water cost based on consumption.

Third, many monitoring systems lack comprehensive cloud-based visualization platforms that allow users to analyze water consumption trends and system status in near real-time.

Finally, there is a lack of integrated IoT systems that combine water level monitoring, water usage measurement, billing calculation, and near real-time cloud visualization in a single framework.

To address these limitations, the proposed system integrates ultrasonic-based water level monitoring, flow sensor-based water usage measurement, near real-time billing estimation, and cloud-based visualization through an IoT dashboard. This integrated approach provides a more efficient and practical solution for smart water management.

III. System Architecture

A. System Overview

The proposed system follows a layered IoT architecture consisting of sensing, processing, communication, and application layers. The ultrasonic sensor measures the water level, while the flow sensor measures water consumption. The ESP32 processes this data and makes decisions regarding pump control and billing. The processed data is transmitted to the Blynk platform for visualization and monitoring. The system leverages a cloud-based IoT platform (Blynk) to ensure remote accessibility, near real-time data visualization, and user interaction.

The block diagram shown in Fig. 1 illustrates the overall architecture of the proposed system. The ultrasonic sensor continuously measures the water level in the tank, while the

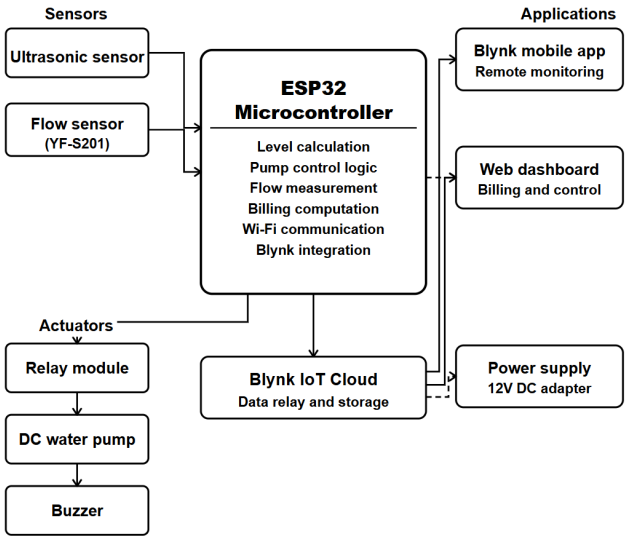

Fig. 1. Block diagram of Smart Water Monitoring System

flow sensor monitors the water consumption by generating pulse signals. These sensor inputs are processed by the ESP32 microcontroller, which acts as the central control unit. Based on predefined threshold conditions, the ESP32 controls the relay module to switch the water pump ON or OFF.

The processed data is transmitted via Wi-Fi to the Blynk IoT cloud platform, where it is visualized in near real-time through a user-friendly dashboard. Additionally, the system activates a buzzer alert when the water level exceeds a critical threshold, ensuring timely user notification.

B. Hardware Components

ESP32 Microcontroller
Acts as the central processing unit of the system. It handles sensor data acquisition, processing, and communication with the cloud platform.

Ultrasonic Sensor
Measures the distance between the sensor and the water surface to determine the tank water level.

Water Flow Sensor (YF-S201)
Generates pulses proportional to water flow, enabling accurate measurement of water usage.

Relay Module
Controls the water pump automatically based on the decisions made by the system.

Buzzer
Provides an audible alert when the tank reaches a critical water level.

Power Supply
Ensures stable voltage and reliable operation for all hardware components.

C. Software and Communication Framework

The system firmware is developed using the Arduino IDE. Communication between the hardware and the cloud platform is established through Wi-Fi connectivity.

- **Platform:** Blynk IoT
- **Protocol:** Wi-Fi
- **Interface:** Mobile and Web Dashboard

D. Data Acquisition and Processing

Water Level Detection
The ultrasonic sensor emits sound waves and measures the return time of the echo to calculate the distance to the water surface.

Water Usage Calculation
The flow sensor produces pulses that are counted using interrupts in the microcontroller.

$$\text{Water Usage (Liters)} = \frac{\text{Pulse Count}}{450} \tag{1}$$

A minimum pulse threshold is applied to reduce noise and improve measurement stability. In this system, pulses below a certain count are ignored to filter out false readings.

E. Billing Calculation

The total water bill is calculated as:

$$\text{Bill} = \text{Total Water Usage} \times \text{Unit Price} \tag{2}$$

In this implementation, a fixed unit price of 0.8 per liter is used for estimation.

The system maintains cumulative records of water usage including:

- Total usage
- Daily usage
- Monthly usage

F. Calibration Approach and Assumption

In practical flow measurement systems, accurate conversion from pulse count to water volume requires experimental calibration of the flow sensor. However, in this implementation, an average calibration factor has been used to simplify the system design and enable rapid prototyping.

The Hall-effect flow sensor (YF-S201) typically produces a pulse frequency proportional to the flow rate. Based on manufacturer specifications and commonly reported values in similar IoT applications, an approximate calibration factor of 450 pulses per liter has been assumed in Eq. (1).

Rationale for Using Average Calibration

- Manufacturer datasheets provide approximate pulse-to-flow relationships
- Flow conditions such as pressure, pipe diameter, and orientation vary in real-world setups
- Experimental calibration requires controlled measurement equipment which was beyond the current checkpoint scope

Therefore, using an average value allows the system to:

- Demonstrate functional billing capability
- Maintain near real-time system performance
- Provide a reasonable estimation of water usage

Impact on System Accuracy

Since the calibration factor is assumed rather than experimentally determined:

- The calculated water usage represents an approximation
- Measurement error may occur depending on flow conditions
- Accuracy may decrease at very low or very high flow rates

However, during testing the system maintained consistent relative measurement, meaning it can effectively track usage trends even if absolute values vary slightly.

Justification in Context of This Work

The primary goal of this system is to demonstrate:

- Integration of flow-based measurement
- near real-time billing computation
- IoT-based visualization

Thus, the use of an average calibration factor represents a practical design trade-off between system simplicity and measurement accuracy.

Display Precision Consideration

Although the system presents water usage values with decimal precision on the dashboard, these values are rounded for better user readability. The underlying measurements are approximate due to the use of an assumed calibration factor, and slight deviations may occur in actual volume estimation.

G. Automation and Actuation

- Pump turns **ON** when the water level is LOW (in automatic mode)
- Pump turns **OFF** when the water level is HIGH
- The buzzer is activated when the water level reaches a critical threshold (near full tank), determined by a predefined distance value.
- A safety lock mechanism is implemented to prevent the pump from restarting immediately after the tank becomes full. The system unlocks only when the water level drops below a safe threshold.
- Users can manually control the pump through the Blynk dashboard, with system-level safety constraints enforced by the controller.

The workflow diagram presented in Fig. 2 describes the operational logic of the system. Initially, the system reads input from both the ultrasonic sensor and the flow sensor. The water level is evaluated against predefined threshold values to determine whether the pump should be activated or deactivated.

Simultaneously, the flow sensor data is processed to calculate water usage using a pulse-to-liter conversion model. The system continuously updates usage statistics and billing information in near real-time. If the water level exceeds a critical limit, the buzzer is triggered to alert the user. All system parameters, including water level, usage, pump status, and billing data, are updated on the IoT dashboard for monitoring and control.

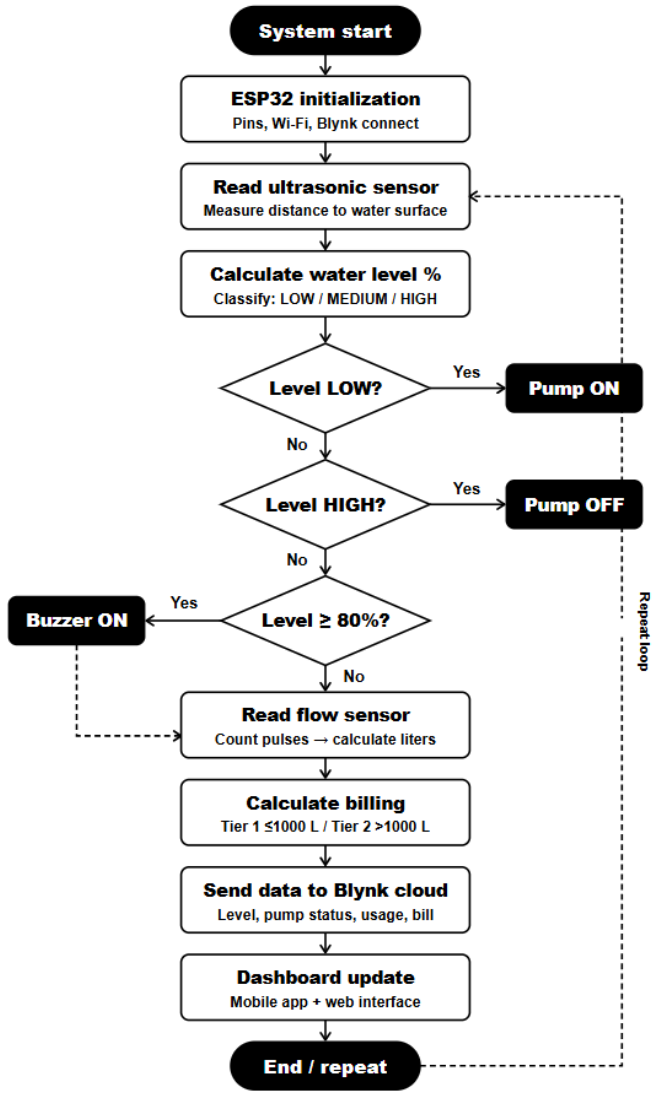

Fig. 2. Workflow diagram of Smart Water Monitoring System

H. Security and Privacy

Basic security mechanisms implemented in the system include:

- Authentication using a Blynk access token
- Secure Wi-Fi communication between the ESP32 and the cloud platform

IV. Prototype Implementation and Experimental Setup

A. Prototype Development

A working prototype of the proposed system was developed using an ESP32 microcontroller, ultrasonic sensor, water flow sensor, relay module, and buzzer. The ESP32 serves as the central processing unit that collects sensor data, processes the information, and communicates with the cloud platform.

The ultrasonic sensor measures the water level in the storage tank, while the flow sensor monitors water consumption by

generating pulses proportional to the flow rate. The relay module controls the water pump automatically based on predefined water level thresholds. Additionally, a buzzer is integrated to provide an audible alert when the water level reaches a critical limit.

The hardware components are integrated with firmware developed using the Arduino IDE. The ESP32 connects to the Blynk IoT cloud platform through Wi-Fi, allowing near real-time monitoring and visualization of system parameters such as water level, water usage, pump status, and billing information.

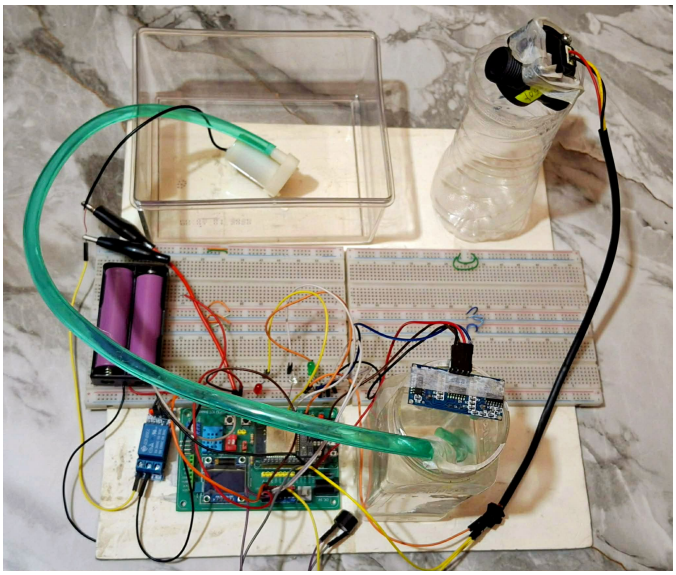

Fig. 3. Hardware Structure of Smart Water Monitoring System

B. Test Environment

The prototype was tested using a small-scale experimental setup consisting of a water tank, a pipeline connected to the flow sensor, and a controllable water supply. Different water levels were simulated by filling and draining the tank while monitoring the response of the ultrasonic sensor.

Controlled water flow conditions were created to evaluate the performance of the flow sensor in measuring water consumption. The relay-controlled pump mechanism and buzzer alert system were also tested under different threshold conditions to ensure correct automation behavior.

The system was connected to the Blynk IoT dashboard to verify near real-time data visualization and remote monitoring capability.

C. Performance Metrics

To evaluate the effectiveness and reliability of the proposed IoT-based smart water monitoring and billing system, several performance metrics were considered. These metrics help assess how well the system operates under practical conditions and whether it satisfies the intended design objectives.

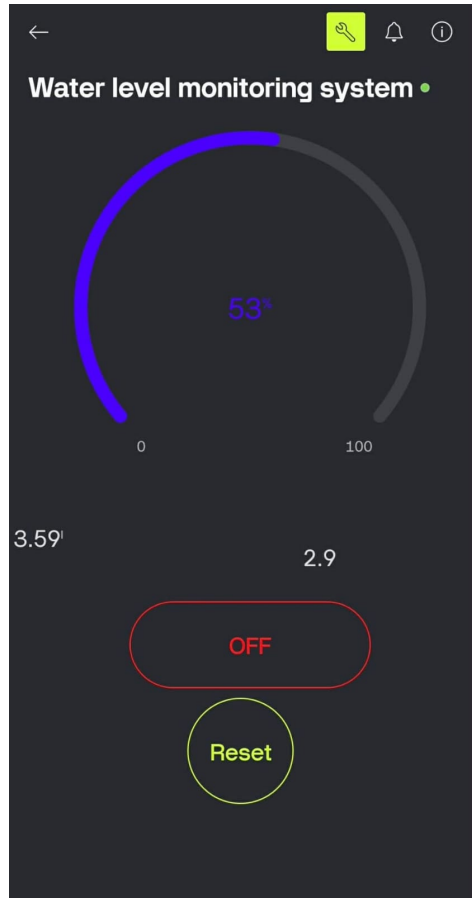

Fig. 4. Water Level Monitoring

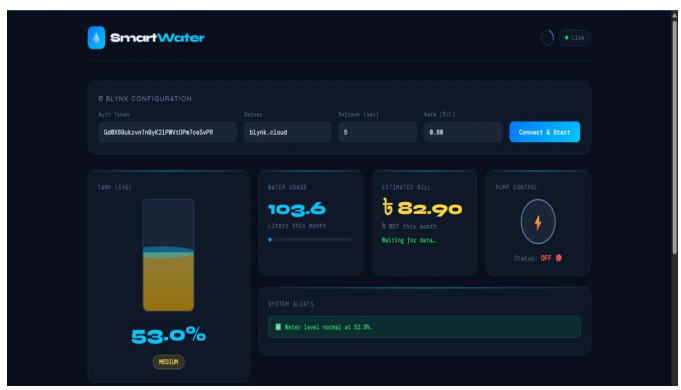

Fig. 5. Dashboad for Billing System

1) Responsiveness: Responsiveness refers to the system's ability to detect changes in sensor inputs and respond within a short period of time. In the proposed system, responsiveness is evaluated based on how quickly the ESP32 processes sensor readings and updates the system outputs.

When the water level in the tank changes, the ultrasonic sensor measures the new distance and sends the data to the ESP32 microcontroller. The microcontroller processes this information and updates the corresponding values on the

Blynk IoT dashboard in near real-time.

Similarly, when water flows through the pipe, the flow sensor generates pulses that are immediately counted using the interrupt mechanism of the ESP32. The system then calculates water usage and updates the billing information accordingly.

High responsiveness ensures that the system can provide near real-time monitoring and control without noticeable delay, which is essential for preventing tank overflow or pump dry-running.

2) Stability: Stability measures the system's ability to operate continuously without crashes, communication failures, or inconsistent behavior. For IoT systems, stability is crucial because the system must maintain long-term connectivity with the cloud platform.

During experimental testing, the proposed system demonstrated stable performance during extended operation periods. The ESP32 maintained consistent communication with the Blynk cloud server, and the sensors continuously provided reliable data streams.

The relay module successfully controlled the pump without unintended switching, while the buzzer alert system activated correctly when the predefined water level threshold was reached.

Stable operation indicates that the system is suitable for real-world deployment where continuous monitoring and automation are required.

3) Functional Accuracy: Functional accuracy refers to how correctly the system performs its intended operations, including water level detection, water usage measurement, pump automation, and billing computation.

The ultrasonic sensor accurately detected variations in water level and triggered pump control actions based on predefined thresholds. The flow sensor generated pulses proportional to the water flow, which were used to estimate water consumption using a pulse-to-liter conversion formula.

Although the system uses an assumed calibration factor of 450 pulses per liter, the measured values remained consistent and proportional to actual water flow during testing. Therefore, the system can effectively track relative water consumption and estimate billing values.

Overall, the experimental results indicate that the system provides reliable monitoring, automation, and billing estimation capabilities.

V. Results and Discussion

A. Experimental Results

The experimental evaluation of the proposed system demonstrated reliable operation across all major components. The following observations were obtained during testing:

- Water level detection remained consistent throughout the testing period.
- Pump automation operated correctly based on predefined water level thresholds.
- The flow sensor generated stable pulse readings corresponding to water flow.
- The billing information was updated dynamically according to measured water usage.

B. Analysis

The experimental results indicate that the proposed system successfully integrates monitoring, automation, and billing within a single IoT-based framework.

Strengths

- near real-time monitoring of water level and usage
- Automated pump control reduces the need for human intervention
- Usage-based billing promotes accountability and efficient water consumption

Limitations

- The assumed calibration factor introduces potential measurement inaccuracies
- Minor fluctuations were observed under very low-flow conditions

The prototype dashboard demonstrates consistent alignment between sensor inputs and computed outputs, with minor deviations attributed to assumed calibration factors and system refresh latency.

VI. Conclusion and Future Work

This paper presents a comprehensive IoT-based water management system that integrates water level monitoring, automated pump control, and flow-based billing. The proposed architecture demonstrates how IoT technology can be used to improve water resource management and provide near real-time insights into water consumption.

A. Conclusion

The developed system successfully demonstrates the integration of sensing, processing, communication, and visualization components within an IoT framework. The results indicate that such systems can significantly improve efficiency in water management while helping reduce water wastage. The proposed system demonstrates a practical pathway toward scalable smart water infrastructure in resource-constrained environments.

B. Future Work

Future enhancements will focus on improving measurement accuracy, system scalability, and intelligent data-driven optimization.

- Implementation of precise flow sensor calibration using experimentally measured water volume
- Development of advanced calibration techniques, including multi-point calibration and adaptive error correction, to improve measurement reliability under varying flow conditions
- Enhancement of the billing system through dynamic tariff models and tier-based pricing strategies
- Integration of multi-user support with account-based usage tracking for shared environments

- Incorporation of a Real-Time Clock (RTC) module for automated billing cycles and monthly usage reset
- Application of data analytics and machine learning techniques to identify consumption patterns and optimize water usage

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